3-Variable & 4-Variable Karnaugh Maps

1. The 3-Variable K-Map

Theory:

A 3-variable K-Map is drawn for a Boolean expression with three variables (e.g., A, B, C). It contains $2^3 = 8$ cells, arranged in a 2×4 grid.

Notice the Gray Code sequence (`00, 01, 11, 10`) along the top axis. This creates a specific numbering layout for the minterms.

3-Variable Grid Layout:

$A \backslash BC$	$00(\bar{B}\bar{C})$	$01(\bar{B}C)$	$11(BC)$	$10(B\bar{C})$
$0(\bar{A})$	m_0	m_1	m_3	m_2
$1(A)$	m_4	m_5	m_7	m_6

(Crucial Note: Remember that the columns jump from 1 to 3, then back to 2. The same happens on the bottom row from 5 to 7, then 6!)

Example 1: 3-Variable SOP

Simplify $F(A, B, C) = \Sigma m(0, 1, 6, 7)$

The Map:

$A \backslash BC$	00	01	11	10
0	**1**	**1**	0	0
1	0	0	**1**	**1**

- **Group 1 (Top Left):** Covers m_0 and m_1. C changes (0 to 1), so it drops. Result: $\bar{A}\bar{B}$.
- **Group 2 (Bottom Right):** Covers m_7 and m_6. C changes (1 to 0), so it drops. Result: AB.
- **Final Expression:** $F = \bar{A}\bar{B} + AB$

Example 2: 3-Variable POS (Grouping 0s)

Simplify $f(A, B, C) = \Pi M(1, 2, 3, 6, 7)$

The Map:

$A \backslash BC$	$00(B + C)$	$01(B + \bar{C})$	$11(\bar{B} + \bar{C})$	$10(\bar{B} + C)$
$0(A)$	1	**0**	**0**	**0**
$1(\bar{A})$	1	1	**0**	**0**

- **Group 1 (Square of four):** Covers M_3, M_2, M_7, M_6. A changes, C changes. Only $\bar{B}$ stays constant. Result: $\bar{B}$.
- **Group 2 (Pair):** Covers M_1, M_3. C changes. Result: $(A + \bar{C})$.
- **Final Expression:** $f = \bar{B} \cdot (A + \bar{C})$

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2. The 4-Variable K-Map

Theory:

A 4-variable K-Map is used for expressions with four variables (e.g., A, B, C, D). It contains $2^4 = 16$ cells, arranged in a 4×4 grid. Gray code is applied to both the vertical and horizontal axes.

4-Variable Grid Layout:

$AB\backslash CD$	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

(Warning: The rows and columns both "skip" at the end. Always double-check where you are placing $m{12}$ through m_{15}!)

Example: 4-Variable SOP with Don't Cares

Simplify $F(A, B, C, D) = \Sigma m(0, 2, 8, 10, 14) + \Sigma d(5, 15)$

The Map:

$AB\backslash CD$	00	01	11	10
00	**1**	0	0	**1**
01	0	**X**	0	0
11	0	0	**X**	**1**
10	**1**	0	0	**1**

Theory - Map Rolling (Wrap-around):

In a 4-variable map, the top row is adjacent to the bottom row, and the left column is adjacent to the right column. You can group "through the walls" of the map.

- **Group 1 (The Four Corners):** m_0, m_2, m_8, m_{10} wrap around all four edges to form a single group of 4. A changes (0 to 1), C changes (0 to 1). We are left with $\bar{B}\bar{D}$.
- **Group 2 (Vertical Pair):** m_{14} and the "Don't Care" m_{10} are grouped. C changes. We are left with $AC\bar{D}$. *(Note: We ignore the Don't Care X_5 because it doesn't help us make*

a bigger group).

- **Final Expression:** $F = \bar{B}\bar{D} + AC\bar{D}$
